电子科技大学
格拉斯哥学院
Glasgow College, UESTC

Physical Experiment II

Prelab Report

Physics Lab 2023

Lab Title: Measure Laser wavelength and index of refraction of air by Michelson interferometer

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Score:

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Answers to Questions (20 points)

1.

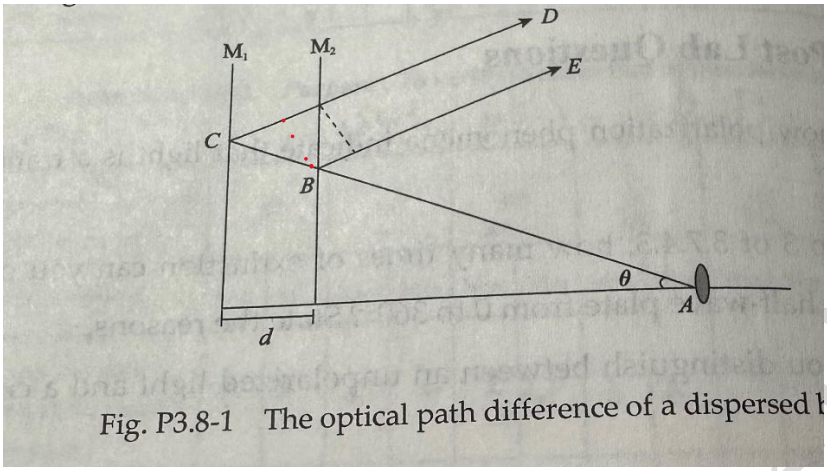


Fig. P3.8-1 The optical path difference of a dispersed l

δ =BC+CM-MN=$(\frac{d}{\cos\theta})$+ $(\frac{d}{\cos\theta})$-2dtan θ sin θ=2dcos θ

2.(1)totally 4 times

(2) To compensate the optical path difference of the two orthogonal polarization modes.

(3)same as G1

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